13 A circle has a diameter AB. The point A has coordinates $(1, -6)$ and the equation of the tangent to the circle at B is $3x + 4y = k$.

(a) Show that the equation of the normal to the circle at the point A is $4x - 3y = 22$. [3]

(b) It is also given that line $x = -1$ touches the circle of the point $(-1, -2)$.
Find the coordinates of the centre and the radius of the circle.
Hence, state the equation of the circle. [5]

(c) Find the value of k. [2]